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COMPUTER MOTHERBOARD

Abstract

The present disclosure provides for a computing system. The computing device may include a motherboard. The motherboard may include a first edge, a second edge, a third edge opposite the first edge, and a fourth edge opposite the second edge. The first edge may be longer than the second edge. The computing device may further include socket for a central processing unit disposed on the motherboard. The computing device may further include an expansion connector. The expansion connector may include one or more expansion slots oriented parallel relative to the motherboard, and be configured to receive an add-on card such that the add-on card is parallel relative to the motherboard.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

[0001] Not applicable.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable.

REFERENCE TO SEQUENCE LISTING OR COMPUTER PROGRAM LISTING APPENDIX

[0003] Not Applicable.

BACKGROUND

[0004] The present disclosure relates generally to computing devices. More particularly, the present disclosure relates to motherboards with add-on cards, such as graphics processing units ("GPUs").

[0005] A personal computer is a stand-alone desktop computing device housed in a chassis. Such computing devices typically include a motherboard and central processing unit ("CPU") positioned thereon, along with other circuitry and electrical components. In many computing devices, an add-on card such as a GPU is connected to the CPU. Conventional computing devices are in alignment with a standard that orients the GPU perpendicular to the motherboard. However, such orientation has been found to present a number of issues. For instance, due to the fact that motherboards are typically oriented parallel to the elongate shape of the chassis, the perpendicular orientation of the GPU relative to the motherboard blocks airflow running through the chassis. Moreover, other components, such as memory modules, are also oriented perpendicular to the motherboard, and thus cut off such airflow that would otherwise reach the GPU itself. As the processing power of computing devices has increased, more heat is generated within computing devices and, thus, more airflow is required in order to cool the GPU (and other components) in order to allow for optimal function of the computing device.

[0006] Additionally, the standard by which many conventional computing devices follow requires a power connector to be oriented perpendicular to the motherboard. Such orientation requires the use of various ribbon cables that must navigate one or more contours in order to reach the correct panel of the chassis, and thus connect to an external power source. Such contours often lead to degradation and otherwise suboptimal function.

[0007] It would be advantageous to provide a computing device that addresses the increasing severity of these issues, in addition to other problems. What is needed, then, are improvements in apparatuses and methods for computing devices.

BRIEF SUMMARY

[0008] This Brief Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0009] One aspect of the present disclosure is a computing device. The computing device may include a motherboard. The motherboard may include a first edge, a second edge, a third edge opposite the first edge, and a fourth edge opposite the second edge. The first edge may be longer than the second edge. The computing device may further include a socket disposed on the motherboard and configured to receive a central processing unit. The computing device may further include an expansion connector. The expansion connector may include one or more expansion slots oriented parallel relative to the motherboard. In some embodiments, the expansion connector is a Peripheral Component Interconnect Express connector.

[0010] In some embodiments, the computing device further includes an add-on card disposed on the expansion connector, such that the add-on card is oriented parallel relative to the motherboard. In further embodiments, the add-on card is coplanar relative to the motherboard. In some cases, the add-on card is a graphics processing unit.

[0011] In some embodiments, the computing device further includes a power connector disposed on the motherboard along the second edge of the motherboard. The power connector may include

one or more power connector slots oriented parallel relative to the motherboard.
[0012] In some embodiments, the computing device further includes one or more input/output connectors disposed on the motherboard located along the fourth edge of the motherboard. In further embodiments, the computing device further includes a memory module disposed on the motherboard.
[0013] In further embodiments, the computing device includes a chassis. The chassis may include a back panel, a sleeve, and a mounting panel. The back panel may be removably secured to the sleeve, and the mounting panel may be disposed on the back panel, such that the mounting panel extends into the sleeve. In such embodiments, the motherboard may be disposed on the mounting panel and oriented parallel to the mounting panel. In some embodiments, the add-on card and the motherboard are oriented vertically. For instance, the add-on card may be located above the motherboard. In this sense, the one or more input/output connectors may be facing the back panel. In some embodiments, the back panel includes an input/output shield aligned with the add-on card.
[0014] Another aspect of the present disclosure is a method of providing the aforementioned computing device. The method may include providing the motherboard, installing the central processing unit on the motherboard, and installing the add-on card on the motherboard, such that the add-on card is oriented parallel to the motherboard.
[0015] In some embodiments, the method may further include installing an expansion connector on the motherboard, securing the motherboard to the back panel, and securing the back panel to the sleeve, such that the motherboard is inserted within the sleeve, and the chassis is formed by the sleeve and the back panel.
[0016] Numerous other objects, advantages and features of the present disclosure will be readily apparent to those of skill in the art upon a review of the following drawings and description of a preferred embodiment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0017] FIG. **1** is a front view of a motherboard of a computing device, according to some embodiments of the present disclosure.
[0018] FIG. **2** is side view of the motherboard of FIG. **1**, according to some embodiments of the present disclosure.
[0019] FIG. **3** is a front view of the motherboard of FIG. **1** with an add-on card connected thereto, according to some embodiments of the present disclosure.
[0020] FIG. **4** is a side view of the motherboard of FIG. **3**, according to some embodiments of the present disclosure.
[0021] FIG. **5** is a front view of an exemplary implementation of a motherboard of a computing device, according to some embodiments of the present disclosure.
[0022] FIG. **6** is a front view of an exemplary implementation of a motherboard of a computing device, according to some embodiments of the present disclosure.
[0023] FIG. **7** is an upper perspective view of the motherboard of FIG. **6**, according to some embodiments of the present disclosure.
[0024] FIG. **8** is a lower perspective view of the motherboard of FIG. **6**, according to some embodiments of the present disclosure.
[0025] FIG. **9** is a rear view of the motherboard of FIG. **6**, according to some embodiments of the present disclosure.
[0026] FIG. **10** is a front view of the motherboard of FIG. **6** with an add-on card connected thereto, according to some embodiments of the present disclosure.
[0027] FIG. **11** is a rear perspective view of the motherboard of FIG. **10** positioned within a

chassis, according to some embodiments of the present disclosure.
[0028] FIG. **12** is a front perspective view of the motherboard of FIG. **10** positioned within a chassis, according to some embodiments of the present disclosure.
[0029] FIG. **13** is a rear view of the motherboard of FIG. **10** positioned within a chassis, according to some embodiments of the present disclosure.
[0030] FIG. **14** is a perspective view of the motherboard of FIG. **10** secured to a back panel for a chassis, according to some embodiments of the present disclosure.
[0031] FIG. **15** is a front view of the motherboard of FIG. **1** with mounting holes, according to some embodiments of the present disclosure.
[0032] FIG. **16** is a schematic view of the motherboard of FIG. **1** with mounting holes, according to some embodiments of the present disclosure.
[0033] FIG. **17** is a rear perspective view of the motherboard of FIG. **14** and a sleeve for a chassis, according to some embodiments of the present disclosure.
[0034] FIG. **18** is a front view of the motherboard of FIG. **14** and a sleeve for a chassis, according to some embodiments of the present disclosure.
[0035] FIG. **19** is a front perspective view of the motherboard of FIG. **14** and a sleeve for a chassis, according to some embodiments of the present disclosure.
[0036] FIG. **20** is a rear perspective view of the motherboard of FIG. **14** and a sleeve for a chassis, according to some embodiments of the present disclosure.
[0037] FIG. **21** is a perspective view of a motherboard for a computing device with a power connector, according to some embodiments of the present disclosure.
[0038] FIG. **22** is a perspective view of a motherboard for a computing device with a power connector, according to alternative embodiments of the present disclosure.

DETAILED DESCRIPTION

[0039] While the making and using of various embodiments of the present invention are discussed in detail below, it should be appreciated that the present invention provides many applicable inventive concepts that are embodied in a wide variety of specific contexts. The specific embodiments discussed herein are merely illustrative of specific ways to make and use the invention and do not delimit the scope of the invention. Those of ordinary skill in the art will recognize numerous equivalents to the specific apparatus and methods described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims.
[0040] In the drawings, not all reference numbers are included in each drawing, for the sake of clarity. In addition, positional terms such as “upper,” “lower,” “side,” “top,” “bottom,” etc. refer to the apparatus when in the orientation shown in the drawing, or as otherwise described. A person of skill in the art will recognize that the apparatus can assume different orientations when in use.
[0041] Referring now to FIGS. **1-2**, a computing device (“device”) **100** is shown, according to some embodiments of the present disclosure. The device **100** may include a motherboard **10** (e.g., a printed circuit board). The motherboard **10** may include a first edge **11** and a second edge **13**. As between the first and second edges **11**, **13**, the first edge **11** may be of a long, straight dimension, and the second edge **13** may be of a short, straight dimension. In other words, the first edge **11** may be longer than the second edge **13**. The motherboard **10** may include a third edge **15** opposite the first edge **11**, and a fourth edge **17** opposite the second edge **13**. The third edge **15** may be of equal (or substantially equal) length relative to the first edge **11**; and the fourth edge **17** may be of equal (or substantially equal) length relative to the second edge **13**.
[0042] As discussed in greater detail below with reference to FIGS. **5-9**, the device **100** may include a socket **33** disposed on the motherboard. The socket **33** may be configured to receive a central processing unit (“CPU”) **24**. It should be appreciated that other such sockets and components received thereon may be disposed on the motherboard **10**, amounting to suitable control circuitry and other electronics. Additionally, as discussed in greater detail below with reference to FIGS. **3-4**, The device **100** may include an add-on card **14**. Generally, the add-on card

14 (e.g., an expansion board, an expansion card, etc.) may be a hardware component that can be connected to the components of the motherboard **10** in order to enhance the functionality of the device **100**. For example, the add-on card **14** may be a graphics processing unit. Of course, the add-on card **14** may be any suitable hardware for being received by an expansion connector **26**, as discussed in greater detail below.

[0043] As shown with reference to FIGS. **3**, and **9**, the motherboard **10** may form a substantially planar body that includes a first face **44** (e.g., a top face on which a substantial amount of such electronics are disposed), and a second face **46** (e.g., a bottom face) opposite and parallel to the first face **44**. As shown with reference to FIG. **4**, the add-on card **14** may form a substantially planar body that includes a first face **66** and a second face **68** opposite and parallel to the first face **66**.

[0044] As mentioned above, the device **100** may include the expansion connector **26**, which may include one or more expansion connector slots **27** configured to interface with a component of the computing device **100**. Depending on the implementation, the expansion connector **26** may be disposed on the motherboard **10** along the first edge **11**. For instance, the expansion connector **26** may be positioned on the first edge **11** such that the expansion connector slots **27** face outwards and away from the motherboard **10** (from a top-view perspective as suggested with reference to FIG. **1**). Of course, the expansion connector slot(s) **27** may be configured in a generally elongate fashion. Thus, when the expansion connector **26** is disposed on the motherboard **26** as shown with particular reference to FIG. **2**, the expansion connector slot(s) **27** may be oriented parallel relative to the motherboard **10** (e.g., parallel relative to the plane generally defined by the planar body of the motherboard **10**).

[0045] The add-on card **14** may be disposed on the expansion connector **26**. Thus, the expansion connector **26** may provide an input/output pathway or route that allows the add-on card **14** to connect to the CPU **24** (and other appropriate components on the motherboard **10**) via printed circuit wiring on the motherboard **10**. Due to the aforementioned orientation of the expansion connector **26**, the add-on card **14**, when disposed on the expansion connector **26** as shown, may be oriented parallel relative to the motherboard **10**.

[0046] In some embodiments, the expansion connector **26** is a peripheral component interconnect ("PCI"). In other embodiments, the expansion connector **26** is a PCI extended ("PCI-X"), which is designed to enhance the legacy PCI for higher bandwidth. In other embodiments still, the expansion connector **26** is a PCI Express ("PCI-e"), which is a high-speed serial computer expansion bus standard designed to replace the legacy PCI and PCI-X. In general, the expansion connector **26** may be a motherboard interface for computer graphics cards, sound cards, hard disk drive host adapters, SSDs, Wi-Fi, and Ethernet hardware connections, and may be configured in any suitable fashion for receiving the add-on card **14** and orienting the add-on card **14** in a parallel configuration relative to the motherboard **10**.

[0047] Referring now to FIGS. **3-4**, the device **100** is shown with the add-on card **14** disposed on the expansion connector **26**, according to some embodiments of the present disclosure. As mentioned above, the expansion connector **26** is disposed on (and is oriented parallel to) the first edge **11**. In this sense, the expansion connector slots **27** may be oriented perpendicularly (or substantially so) relative to the first edge **11**. Thus, the add-on card **14** (or, more generally, the planar feature broadly defined by the shape of the add-on card **14**) may be oriented in parallel relative to the motherboard **10** (or, more generally, the planar feature broadly defined by the shape of the motherboard **10**). As a first example, the first and/or second faces **44**, **46** of the motherboard **10** may be parallel to the first and/or second faces **66**, **68** of the add-on card **14**. As a second example, a motherboard axis **62** (e.g., an axis that runs parallel to the first and/or second faces **44**, **46** of the motherboard **10**) may be parallel to a GPU axis **64** (e.g., an axis that runs parallel to the first and/or second faces **66**, **68** of the add-on card **14**. In further embodiments, the add-on card **14** is coplanar with the motherboard **10**.

[0048] Referring now to FIGS. **5-9**, the device **100** is shown, according to some embodiments of the present disclosure. As mentioned above, the device **100** may include the CPU **24** disposed on the motherboard **10**, along with control circuitry and other electronics. As a first example, device **100** may include a memory module connector **40** disposed on the motherboard **10**. For instance, the memory module **40** may be a random access memory ("RAM") slot **40**. In this sense, and as depicted with reference to FIG. **11**, the device **100** may include a memory card **41** disposed on the memory module **40**. As a second example, the device **100** may include input/output ("IO") connectors **38**, which may facilitate the connections of IO devices such as a video monitor, mouse, or printer to the control circuitry and other electronics of the device **100**. In general, and as depicted with particular reference to FIG. **5** and with additional reference to FIG. **10**, the motherboard **10** and one or more of the aforementioned components disposed thereon or connected thereto, including the add-on card **14**, may form a motherboard assembly **90**.

[0049] Referring now to FIG. **10**, the device **100** is shown with the motherboard assembly **90** secured to a chassis **12** (or a portion of thereof, as discussed in greater detail below), according to some embodiments of the present disclosure. As shown, the add-on card **14** and the motherboard **10** may be oriented vertically. Thus, the add-on card **14** may be located above the motherboard **10**. Of course, in further embodiments, the orientation may be adjusted such that the add-on card **14** is located below the motherboard **10**. Moreover, it should be appreciated that the computing system **100** may be oriented in any number of suitable fashions, such that the add-on card **14** is located to the left or right of the motherboard **10**. Indeed, it should be noted that the descriptions and depictions discussed herein are exemplary in nature, and should be considered non-limiting embodiments of the present disclosure.

[0050] As discussed in greater detail below with reference to FIGS. **11-13**, the device **100** may include the chassis **12** configured to house and secure the motherboard assembly **90**. The chassis **12** may include a back panel **18**. The motherboard **10** and/or the add-on card **14** may be mounted on or otherwise secured to the back panel **18**. As shown, the motherboard **10** and/or the add-on card **14** may be oriented perpendicular to the back panel **18**.

[0051] In some embodiments, the back panel **18** includes an interface **20** configured to secure the one or more IO connectors **38** disposed on the motherboard **10**. Additionally, the back panel **18** may include an IO shield **23**. The IO shield **23** may be aligned with the add-on card **14**. The IO shield **23** may be configured to secure additional IO connectors disposed on the add-on card **14**. In some embodiments, particularly in cases where the add-on card **14** is a GPU, the IO shield **23** may include a vent. While the position of the IO shield **23** may be moved to any suitable location(s) on the chassis **12**, it should be appreciated that the parallel position of the add-on card **14** relative to the motherboard **10** may advantageously allow a free stream of air to pass through the chassis **12**, thus enhancing the cooling effects resulting from air travel throughout the chassis. Moreover, the chassis **12** may be reduced in terms of footprint size due to the vertically structured arrangement of the motherboard assembly **90**.

[0052] Referring now to FIGS. **11-13**, the device **100** is shown with the motherboard assembly **90** housed within the chassis **12**, according to some embodiments of the present disclosure. In addition to the back panel **18** (e.g., a back panel), the chassis **12** may further include sleeve **19**. The sleeve **19** may be formed as an open box by a front panel **48**, a top panel **50**, a bottom panel **52**, a right panel **54**, and a left panel **56**. The back panel **18** may be removably secured to (e.g., over the open portion of) the sleeve **19**. When the back panel **18** is secured to the sleeve **19** as shown, the chassis **12** may form a box as suggested above. Owing to the aforementioned box shape, the back and front panels **18**, **48** may be parallel (or substantially parallel), the top and bottom panels **50**, **52** may be parallel (or substantially parallel), and the left and right panels **54**, **56** may be parallel (or substantially parallel). Thus, the chassis **12** may form a box around the motherboard assembly **90**.

[0053] Referring now to FIG. **14**, the device **100** may include a removable motherboard insert **25**, according to some embodiments of the present disclosure. For instance, the removable motherboard insert **25** may include the motherboard assembly **90** depicted with reference to FIGS. **5** and **9**, a mounting panel **22**, and, as suggested above, the back panel **18**. Of course, the mounting panel **22** may be considered an internal component of the chassis **12**. Thus, the chassis **12** may include the back pabel **18**, the sleeve **19**, and the mounting panel **22**. Such components may be secured to one another in any suitable combination. As a first example, the motherboard assembly **90** may be secured to the back panel **18** as depicted with reference to FIG. **10**, and the back panel **18** may be secured to the mounting panel **22**. As a second example, the motherboard assembly **90** may be secured to each of the mounting panel **22** and the back panel **18**.

[0054] Referring now to FIG. **15**, the motherboard assembly **90** may be secured to the mounting panel **22** via a number of mounting holes, according to some embodiments of the present disclosure. For instance, the motherboard **10** may include a first mounting hole **28**, a second mounting hole **30**, a third mounting hole **32**, a fourth mounting hole **34**, and a fifth mounting hole **36**. As shown, the first, second, and third hole **28**, **30**, and **32** may be located along the first edge **11**, while the fourth and fifth holes **34**, **36** may be located along the third edge **15**.

[0055] Referring now to FIG. **16**, dimensions of the motherboard **10** are shown, according to some embodiments of the present disclosure. In some embodiments, a distance L**1** between the first and third edges **11**, **13** (e.g., the general span of the second and fourth edges **13**, **17**) is 9.6 inches. In some embodiments, a distance H**1** between the second and fourth edges **13**, **17** (e.g., the general span of the first and third edges **11**, **13**) is 10.78″. In other embodiments, the distance H**1** is 6.7 inches.

[0056] The first, second, and third mounting holes **28**, **30**, **32** may be horizontally aligned (e.g., aligned on an axis parallel to the first edge **11**) and vertically offset from the first edge **11** by a length H**6** of 0.25 inches. The fourth and fifth mounting hole **34**, **36** may be horizontally aligned and vertically offset from the third edge **15** by H**6**.

[0057] The first, second, and third mounting hole **28**, **30**, **32** may be positioned above (e.g., closer to the expansion connector **26** relative to the fourth edge **17**) the fourth and fifth mounting hole **34**, **36**, and be vertically offset from the fourth and fifth mounting hole **34**, **36** by a distance H**2** of 6.2 inches.

[0058] The first mounting hole **28** may be horizontally offset from the second mounting hole **30** by a length L**2** of 5.2 inches. The third mounting hole **32** may be horizontally offset from the second mounting hole **30**, where the second mounting hole **30** the middle mounting hole amongst the first, second, and third mounting hole **28**, **30**, **32**, by a length L**3** of 2.85 inches.

[0059] In some embodiments, the first mounting hole **28** may be horizontally offset from the fourth edge **17** by a length L**5** of 1.5 inches. In particular, the length L**5** may represent a distance between the fourth edge **17** and the nearest edge of the expansion connector **26**. As suggested above with reference to FIG. **10**, additional IO connectors may disposed on the add-on card **14** (and secured by the IO shield **23**). Such additional IO connectors may generally be vertically aligned with the forth edge **17**. In this sense, such IO connectors may necessitate a horizontal gap between the fourth edge **17** and the nearest edge of the add-on card **14**. Thus, the length L**5** (as with any of the dimensions discussed herein) may be any dimension suitable for facilitating the apparatuses and methods discussed herein.

[0060] The fourth mounting hole **34** may be horizontally offset from the fourth edge **17** by a length L**6** of 0.4 inches, and be positioned to the left of the fifth mounting hole **36**. The fourth mounting hole **34** may be horizontally offset from the fifth mounting hole **36** by a length of 5.8 inches.

[0061] The IO connectors **38** may be arranged such that they are generally positioned on a rectangular region that is lower than, and vertically offset from the first, second, and third mounting hole **28**, **30**, **32** by a length H**4** of 0.4 inches. The external connectors may be higher than, and vertically offset from the fourth and fifth mounting holes **34**, **36** by a length H**5** of 0.3 inches. The external connectors may be arranged such that, horizontally, the are generally aligned with the fourth edge **17** and extend onto the motherboard **10** by a length L**4** of 1.5 inches.

[0062] Of course, as suggested above with reference to the length L**5**, the various dimensions described herein are merely exemplary in nature. It should be appreciated that the dimensions of the motherboard **10**; the first, second, third, fourth, and fifth mounting holes **28**, **30**, **32**, **34**, **36**; the IO connectors **38**; and any other component of the device **100** discussed herein may be dimensioned or arranged in any suitable format in order to provide the advantages discussed herein.
[0063] Referring now to FIGS. **17-20**, the device **100** is shown to depict the removal of the removable motherboard insert **25** from, or insertion of the same into, the sleeve **19**, according to some embodiments of the present disclosure. As suggested above, the sleeve **19** may be arranged such that an opening **60** is left where the back panel **18** would otherwise be. The dimensions of the removable motherboard insert **25** may be arranged such that the removable motherboard insert **25** is insertable within the sleeve **19**. Generally, this advantageously provides access to the motherboard **10** in a manner that may not require significant steps or manual labor, such as disassembling a side panel, as may be the case in conventional computing devices.
[0064] Referring now to FIGS. **20** and **21**, the device **100** is shown, according to some embodiments of the present disclosure. The device **100** may include a power connector **42**. The power connector **42** may be disposed on the motherboard **10** along the second edge **13**, and include one or more power connector slots **43** oriented parallel relative to the motherboard **10** (similar to the discussion of the expansion connector **26** above). The power connector **42** may face in an opposite direction relative to the one or more IO connectors **38**, thus facing the front panel **48** of the chassis. By orienting the power connector **42** as such, any cabling required to bridge between the power connector **42** and an external power supply may not be required to undergo as many contours that are typically required in conventional computing systems. Depending on the implementation, the power connector **42** may be disposed on the first (e.g., top) face **44** of the motherboard **10** (as shown with reference to FIG. **20**), the second (e.g., bottom) face **46** of the motherboard **10**, or in some other suitable configuration.
[0065] Thus, although there have been described particular embodiments of the present invention of a new and useful COMPUTER MOTHERBOARD, it is not intended that such references to particular embodiments be construed as limitations upon the scope of this invention.

Claims

1. A computing device, the device comprising: a motherboard including a first edge, a second edge, a third edge opposite the first edge, and a fourth edge opposite the second edge, wherein the first edge is longer than the second edge; a socket disposed on the motherboard and configured to receive a central processing unit; and an expansion connector disposed on the motherboard along the first edge, the expansion connector including one or more expansion connector slots oriented parallel relative to the motherboard.
2. The device of claim 1, further comprising an add-on card disposed on the expansion connector, such the graphics processing unit is oriented parallel relative to the motherboard.
3. The device of claim 2, wherein the add-on card is coplanar relative to the motherboard.
4. The device of claim 3, wherein the expansion connector is a Peripheral Component Interconnect Express connector.
5. The device of claim 4, wherein the add-on card is a graphics processing unit.
6. The device of claim 5, further comprising a power connector disposed on the motherboard along the second edge, the power connector including one or more power connector slots oriented parallel relative to the motherboard.
7. The device of claim 5, further comprising one or more input/output connectors disposed on the motherboard and located along the fourth edge.
8. A computing device, the device comprising: a chassis including a back panel, a sleeve, and a mounting panel, wherein the back panel is removably secured to the sleeve and the mounting panel

is disposed on the back panel and extends into the sleeve; a motherboard disposed on the mounting panel and oriented parallel to the mounting panel; a socket disposed on the motherboard, the socket configured to receive a central processing unit; and an expansion connector disposed on the motherboard, the expansion connector including one or more expansion slots oriented parallel relative to the motherboard.

9. The computing device of claim 8, wherein the expansion slots and the motherboard are oriented vertically.

10. The computing device of claim 9, further comprising an add-on card disposed on the expansion connector, the add-on card oriented vertically.

11. The computing device of claim 10, further comprising one or more input/output connectors disposed on the motherboard and facing the back panel.

12. The computing device of claim 11, wherein the back panel includes an input/output shield aligned with the add-on card.

13. The computing device of claim 8, wherein the add-on card is located above the motherboard.

14. The computing device of claim 8, wherein the add-in card is a graphics processing unit.

15. A method of providing a computing device, the method comprising: providing a motherboard including a first edge, a second edge, a third edge opposite the first edge, and a fourth edge opposite the second edge, wherein the first edge is longer than the second edge; installing a central processing unit on the motherboard; and installing an add-on card on the motherboard, such that the add-on card is oriented parallel to the motherboard.

16. The method of claim 15, further comprising installing an expansion connector along the first edge of the motherboard, such that one or more expansion connector slots of the expansion connector are oriented perpendicular to the motherboard, wherein the add-on card is disposed on the expansion connector slots.

17. The method of claim 15, further comprising: securing the motherboard to a back panel; and securing the back panel to a sleeve, such that the motherboard is inserted within the sleeve, and a chassis is formed by the sleeve and the back panel.

18. The method of claim 17, further comprising installing one or more input/output connectors along the fourth edge of the motherboard, such that the one or more input/output connectors face the back panel.

19. The method of claim 18, further comprising installing one a power connector along the second edge of the motherboard, such that the power connector faces in an opposite direction relative to the one or more input/output connectors.

20. The method of claim 19, wherein the add-on card is located above the motherboard.